EVERY QCQS SPACE WITH FINITELY MANY IRREDUCIBLE COMPONENTS BECOMES RESOLUTION-PROPERTY AFTER A DENSE ADMISSIBLE BLOW-UP

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 13832-13839 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 13841-13878, the complete author proof (36 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Capstone of the spaces chapter's resolution-property series; the parenthetical (details omitted) covers only the harmless shrinking f^{-1}(U) subset V after shrinking U. Cross-chapter dependency on card L16087 is now satisfied (it assembled this morning).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The chapter's finite-refinement limit theorem (Limits of Spaces), the schemes-side blow-up-ample-family theorem (card L16087, now on the shelf), the strict-transform finiteness lemma (this chapter's lemma finite-after-blowing-up, same wave L16118), projectivity of strict transforms (Divisors on Spaces), ample-family-with-relative-ample machinery (More on Morphisms), and the two resolution-property transfers (Derived Categories of Schemes/Spaces).

Source reference. Original lemma 0B4K: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 8913–8921. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GTP: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13878–13878. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GTT: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13878–13878. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GMM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13878–13878. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0864: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13878–13878. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GUN: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13878–13878. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GUV: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 13878–13878. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Assume that $|X|$ has finitely many irreducible components. There exists a dense quasi-compact open $U \subset X$ and a U-admissible blowing up $X' \to X$ such that the algebraic space X' has the resolution property.*

Proof. By Limits of Spaces, Lemma 0GUN there exists a surjective, finite, and finitely presented morphism $f : Y \to X$ where Y is a scheme and a quasi-compact dense open $U \subset X$ such that $f^{-1}(U) \to U$ is finite étale. By More on Morphisms, Lemma 0GTT there is a quasi-compact dense open $V \subset Y$ and a V-admissible blowing up $Y' \to Y$ such that Y' has an ample family of invertible modules. After shrinking U we may assume that $f^{-1}(U) \subset V$ (details omitted). Hence $f' : Y' \to X$ is finite étale over U and in particular, the morphism $(f')^{-1}(U) \to U$ is finite locally free. By Lemma 0B4K there is a U-admissible blowing up $X' \to X$ such that the strict transform Y'' of Y' is finite locally free over X'. Picture

$$\begin{array}{ccccc} Y'' & \xrightarrow[g]{} & Y' & \longrightarrow & Y \\ \downarrow & & & & \downarrow \\ X' & & \longrightarrow & & X \end{array}$$

Since $g : Y'' \to Y'$ is a blowing up (Divisors on Spaces, Lemma 0864) in the inverse image of the center of $X' \to X$, we see that $g : Y'' \to Y'$ is projective and that there exists some g-ample invertible module on Y''. Hence by More on Morphisms, Lemma 0GTP we see that Y'' has an ample family of invertible modules. Hence Y'' has the resolution property, see Derived Categories of Schemes, Lemma 0GMM. We conclude that X' has the resolution property by Derived Categories of Spaces, Lemma 0GUV. □